IBN SINA UNIVERSITY

FIRST YEAR

GENERAL CHEMISTRY

Definition

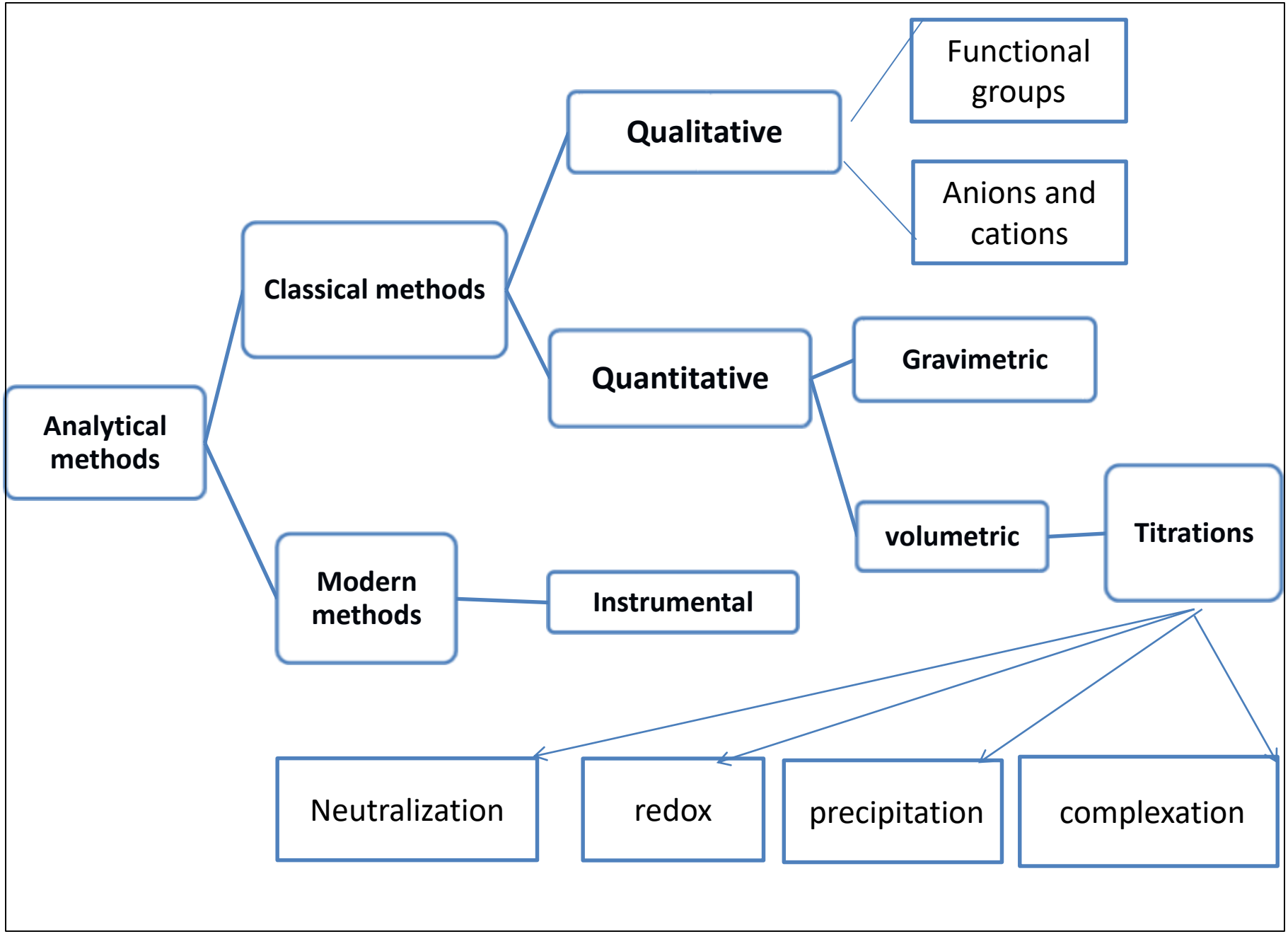
- **Chemistry** is a branch of science that studies the composition, structure, properties and change of matter

Analytical Chemistry

- **Analytical Chemistry:** The science seeks ever improved means of measuring the chemical composition of natural and artificial materials by using techniques to identify the substances which may be present in a material and to determine the exact amounts of the identified substance.

Analytical chemistry

involves the analysis of matter to determine its composition and the quantity of each kind of matter that is present. Analytical chemists detect traces of toxic chemicals in water and air. They also develop methods to analyze human body fluids for drugs, poisons, and levels of medication.


Analytical methods
Classical methods
Modern methods
Qualitative
Quantitative
Functional groups
Anions and cations
Gravimetric
volumetric
Titrations
Instrumental
Neutralization
redox
precipitation
complexation

Analytical chemistry consists of:

(A) Qualitative analysis which deals with the identification of elements, ions, or compounds present in a sample (tells us what chemicals are present in a sample).

(B) Quantitative analysis which is dealing with the determination of how much of one or more constituents is present (tells how much amounts of chemicals are present in a sample). This analysis can be divided into **three branches**:

(1)Volumetric analysis (Titrimetric analysis): The analyte reacts with a measured volume of reagent of known concentration, in a process called titration.

(2) Gravimetric analysis: usually involves the selective separation of the analyte by precipitation, followed by the very non-selective measurement of mass (of the precipitate).

(3) Instrumental analysis: They are based on the measurement of a physical property of the sample for example atomic spectroscopy , mass spectrometry, nuclear magnetic resonance spectrometry (NMR), X-ray spectroscopy

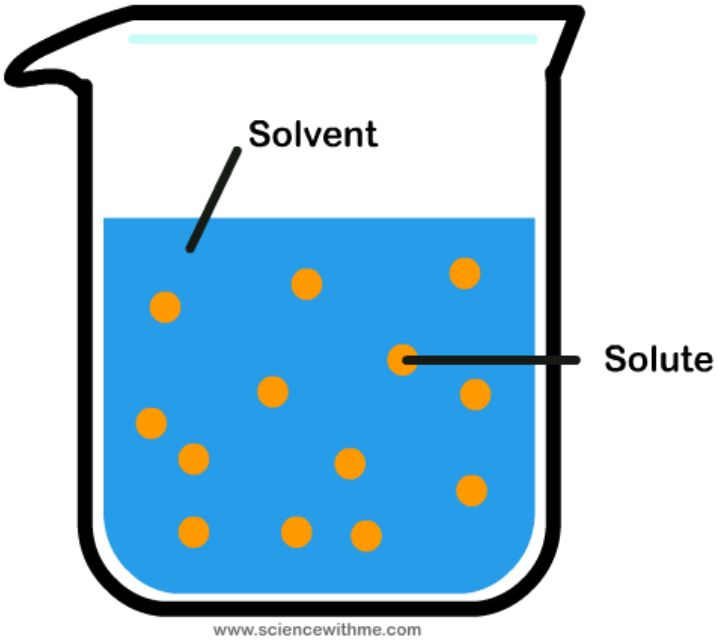
A- The Solutions:

A solution is a homogeneous mixture of two or more substances.

-The solute is the substance present in a smaller amount,

-The solvent is the substance present in a larger amount.

***Homogeneous* means that the components of the mixture form a single phase.**



Solution

-Aqueous Solution:

In aqueous solutions,
the solute is a liquid or solid &
the solvent is water.

Making aqueous solution by dissolving table salt (NaCl) in water.

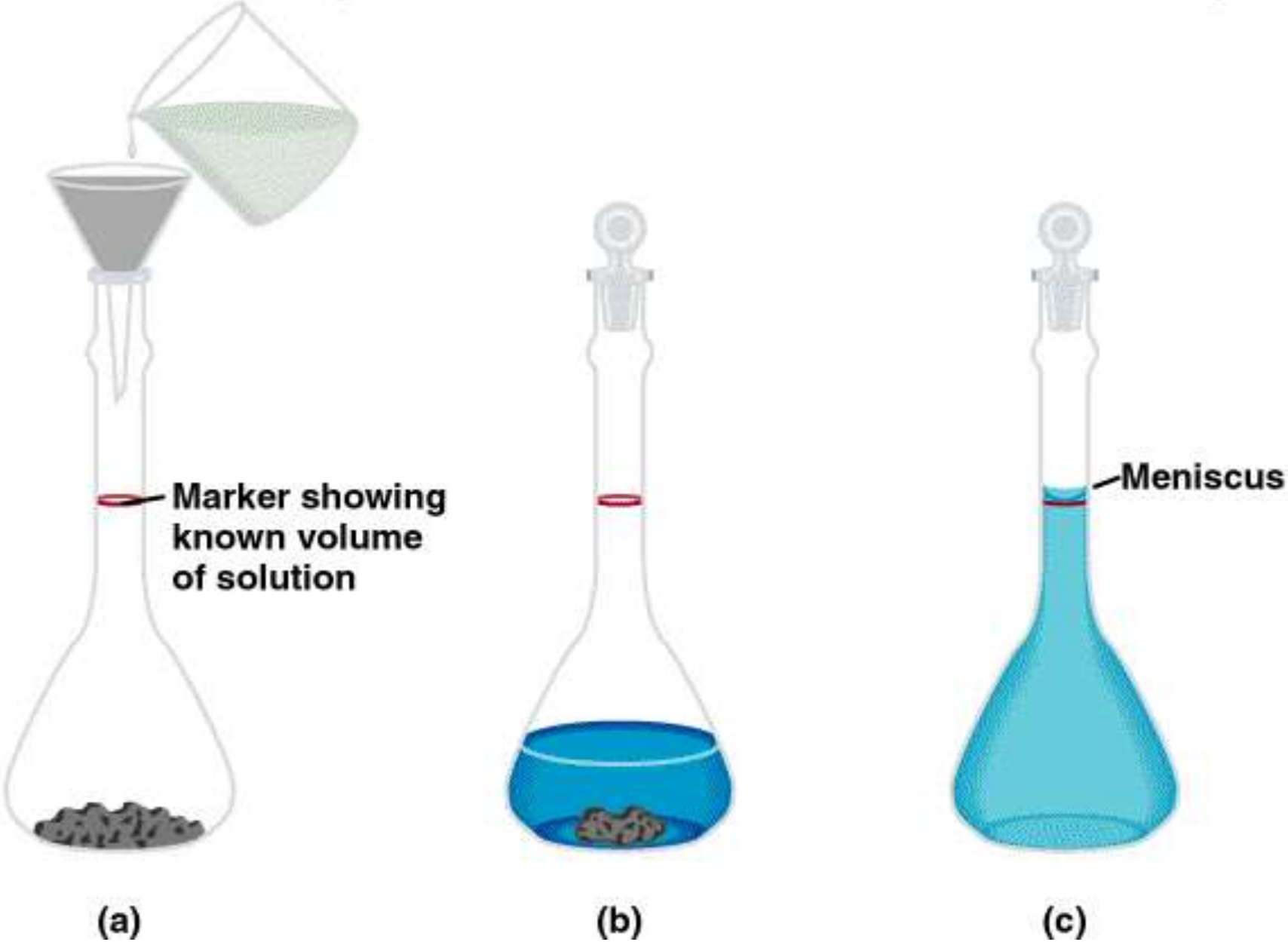
The salt is the solute and the water the solvent.

Preparing a solution of known Concentration.

In titrations the concentration of one of the solutions must be known(standard solution). The standard solution is prepared by taking a certain weight of a chemical and dissolving it in a water to give a solution whose volume is accurately known. The known solution is usually made up in a volumetric (or graduated) flask.

VOLUMETRIC FLASKS

Preparing a Solution of Known Molarity



If we wish to dissolve a solid in water to give a solution of known concentration, the first step we have to know two things:

- **The mass of the solid we use.**
- **The total volume of the solution.**

Example:

If We might want to make a solution of sodium carbonate that has 5.3 g of the solid in 250 ml of solution. It is tempting to think that all we have to do is fill the volumetric flask to the graduation mark, weigh out the sodium carbonate and then put it in the flask……..?

The problem of doing this is that the final volume of the solution would not be 250 ml.

when solids dissolve in water, there is nearly always a volume change.

The correct method is to dissolve the solid in water (less than 250 ml), first add this to the flask and top up with enough water to reach the graduation mark.

Standard solution:

Standard solution is a solution of accurately known concentration.

Standard solutions are usually prepared from primary standard substances.

primary standard substances are certain substances which have characteristic properties which differentiate them from other chemicals

Characteristic properties of primary standard substances:

1-Easily available in high degree of purity.

2-Completely stable and unaffected by atmosphere.

3-Easily dissolved in the desired solvent.

4. Should have large molecular weight to minimize the weighing errors.

and their solutions (primary standard solutions) should not decompose on keeping

Saturated & Unsaturated solution

Suppose that you have a beaker of water to which you add some salt, stirring until it dissolves. So you add more and that dissolves. You keep adding more and more salt, eventually reaching a point that no more of the salt will dissolve no matter how long or how vigorously you stir it because we get a saturated state to the water salt solution which we make it .

A saturated solution: *is a solution that contains the maximum amount of solute that is capable of being dissolved. At 20° C, If any more of solute is added it will not dissolve because the solution is saturated.*

An un saturated solution: *is a solution that is less concentrated (more dilute) than a saturated solution.*

The mole

Is the SI base unit of amount of atoms, ions, or molecules (particles).

Mole:

Mole is the mass of substance that equals the atomic or molecular mass of that substance.

** One mole of any substance contains Avogadro's number of atoms ,molecules, or ions.

Avogadro's number, (N_A) = 6.022×10^{23}

Number of atoms, molecules, or ions = n x N_A

(N_A) = Avogadro's number,
(n) = Number of moles

Using this constant we can calculate the number of atoms in any quantity of moles.

Example, the number of atoms in 3 mole C-12 is 6.022×10^{23} mole $^{-1}$ x 3 mole = 18.066×10^{23} atoms.

The molecular weight,

the **molecular weight (M.wt),** is the weight of all atoms in a given molecular formula . Molecular weight is measured in Atomic Mass Units

Molecular weight =
∑((Atomic Mass of Element)x (N of atoms of that element)

Find the molecular weight of methane (CH_4).
The atomic mass of hydrogen is 1 The atomic mass of carbon is 12 . (C =12 , H =1)
Solution : M.wt = **[(1 * 12) + (4*1)]** = 16

Find the molecular weight of water (H_2O).

The atomic mass of oxygen is 16 ,The atomic mass of hydrogen is 1 . (O =16 , H =1)

Solution : M.wt = **[(2*1) + (1*16)]** = 18

Calculate the molecular weight of glucose ($C_6H_{12}O_6$).

(C = 12 , O = 16 , H = 1)

M.wt = [(6*12) + (12*1) + (6*16)]

M.wt = [72 + 12 + 96]

M.wt = 180

Some element such as Hydrogen, chlorine Nitrogen, Exist as diatomic(molecules)

H_2 , Cl_2 , N_2 .

*****So if we had 2 moles of N_2 we would have 4 mole of atoms.**

Example :

Calculate the total number of atoms in 0.64 mol of ozone gas O3

Solution :-

we had 0.64 moles of O3 we would have

(3*0.64) = 1.92mole

Number of atoms = 1.92* 6.022×10^{23} = 1.15×10^{24} atoms

$$\text{No. of } \textbf{moles} \text{ of a substance} = \frac{\text{mass (weight) of substance}}{\text{its molar mass (molecular weight)}}$$

- Example: A bottle contain 6 g of magnesium ribbon, how many moles of magnesium present?.
- $$\text{No. of moles of Mg} = \frac{\text{mass of Mg}}{\text{Atomic mass of Mg}}$$
$$= 6/24 = 0.25 \text{ mole.}$$

Example :

Calculate the number of moles and the total number of atoms in the following:

A. 5.06 grams of oxygen (O_2)

solu,. Number of moles =5.06/ 32=0.158 moles

Number of molecules = $0.158 \times 6.022 \times 10^{23} = 9.51 \times 10^{22}$

Atoms per molecule 2 (since O_2)

Total number of atoms = 9.51×10^{22} molecules x 2 atoms= 1.902×10^{23}

B . 2.14 grams of potassium (K) (Atomic weight K = 39.1)

Example

*How many atoms would there be in: (**i) 4 mole of Si***

(ii) 0.2 mole of Br_2

Home work(1)

1. *How many atoms would there be in: (i) 0.5 mole of Na, (ii) 128 grams of cupper Cu (iii) 5 mole of neon gas Ne, (iv) 56 g of nitrogen* N_2
2. *What the mass of : (i) 2 mole of calcium, Ca (ii) 1 mole of phosphine,* PH_3 *(iii) 0.25 mole of silver nitrate,* $AgNO_3$.
3. *Work out the number of moles in each of the following: 2 g of methane,* CH_4 *(ii) 140 g of calcium oxide, CaO, (iii) 25 gof cupper (II) sulphate crystals,* $CuSO_4.5H_2O$.

Expressing concentrations

The concentration of solution refers to the weight or volume of the solute present in specified amount of the solvent or solution.

Percent Composition:

One of the easiest methods of making a mixture with a specific composition is to measure the masses of each of the parts and express the concentration as a percentage of solute to the mass of the entire mixture.

There are several common methods of expressing concentrations:

- Percent by weight: the percentage given refers to the solute.

- Percent by volume: the concentration of a solution of two liquids.

$$\text{Mass \%} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\ \%$$

$$\text{Volume \%} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\ \%$$

Examples:

1- As an example consider 5 g sugar dissolved in 20 g of water. What is the w/w% concentration of sugar in this solution?

$$\frac{5\text{ g sugar}}{25\text{ g solution}} \times 100 = 20\text{ w/w }\%$$

2- As an example consider 50 ml of Ethanol dissolved in 100 ml of water. What is the V/V% concentration of Ethanol in this solution?

$$\frac{50\text{ ml Ethanol}}{150\text{ ml Solution}} \times 100 = 33.3\%$$

Home work(2):

1- A solution is prepared by mixing 5 grams of ethanol (C_2H_5OH) with 100 grams of water . Calculate the mass percent of ethanol in this solution? ***(mass of solution = mass of solute + mass of solvent)***

2- A 135-g sample of sea water is evaporated to dryness leaving 4.73g of solid residue (the salts formerly dissolved in sea water). Calculate the mass percent of solute present in the original sea water.

3- Cows milk typically contains 4.5% by mass of sugar lactose, $C_{12}H_{22}O_{11}$. Calculate the mass of lactose present in 175g of milk.

Similar unit that is used to express very small concentrations (e.g., the concentration of impurities, pollutant in air or water) is parts per million, (ppm) by volume or by weight.

Concentration in parts per thousand or million or billion:-

$$ppt = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 10^3 = \frac{\text{wt (mg)}}{\text{wt (g)}} = \frac{\text{wt (g)}}{\text{wt (kg)}}$$

$$Ppm = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 10^6 = \frac{\text{wt (µg)}}{\text{wt(g)}} = \frac{\text{wt (mg)}}{\text{wt (kg)}}$$

$$Ppb = \frac{\text{mass of solute g}}{\text{mass of solution g}} \times 10^9 = \frac{\text{wt (n g)}}{\text{wt (g)}} = \frac{\text{wt (µg)}}{\text{wt (kg)}}$$

• Parts Per Million

Symbol: ppm

Definition: It is defined as the parts of a component per million parts (10^6) of the solution. It is widely used when a solute is present in trace quantities.

Unit: ppm

Example:

0.02 gram of NaCl was dissolved in 1000 grams of solution, calculate the concentration of this solution, in ppm?

solution:

Weight of NaCl = 0.02 g Weight of the solution = 1000 g

$$\text{Conc. in ppm} = \frac{\text{Wt (g)}}{\text{Wt (g)}} = \frac{0.02 \times 1{,}000{,}000}{1000} = 20 \text{ ppm}$$

Mole fraction:

- The mole fraction of component A in a solution is given by (XA).
- XA = number of moles of A/total number of moles

$$\text{Mole Fraction of Solute, } X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

$$\text{Mole fraction of } X_{solvent} = \frac{n_{solvent}}{n_{solvent} + n_{solute}}$$

Where $X_{solvent} + X_{solute} = 1$

- Example:

- Calculate the mole fraction of ethyl alcohol C_2H_5OH& water,H_2O in a solution prepared by adding 92 grams of ethyl alcohol to 108 grams of water.

.To a component in a solution the mol fraction is *less than one*

.The sum of the mol fractions is *equal one*

Home work (3)
mole fraction

- For the following solution find the mole fraction of both the solute and solvent
- 1- 2 mole of vinyl methyl ether and 6 moles of water
- 2-117grams of NaCl dissolved in 360 grams of water.

Molarity (M): molarity is widely used in talking about concentration in chemistry, (IUPAC) system.

Definition: The number of moles of solute per liter of solution

By definition:

- Molarity (M) = $\frac{\text{no. of moles of solute}}{\text{volume (in liters) of solution}}$ .....(1)
- M = $\frac{\text{wt}}{\text{M.wt x volume (in liters)}}$ .....(2)
- M = $\frac{\text{wt x 1000}}{\text{M.wt x V (Cm}^3\text{)}}$ .....(3)
- Wt = $\frac{\text{M x M.wt x V(Cm}^3\text{)}}{1000}$ .....(4)

From equation (1):

- No. of moles = $\frac{M \times V(Cm^3)}{1000}$ .....(5)

In dilutions: $\frac{M1 \times V1}{1000} = \frac{M2 \times V2}{1000}$ .....(6)

- $M1 \times V1 = M2 \times V2$ .....(7)

In neutralization reactions:

- $\frac{Ma \times Va}{a} = \frac{Mb \times Vb}{b}$

Example : Calculate the molarity of a solution made by dissolving 1.0 g of phenol C_6H_5OH, in enough water to make 50 ml of solution.

Solution: no. of moles = $\frac{wt}{M.wt}$ = 1.0/94=0.01 mole

- M = $\frac{\text{no. of moles}}{V\ (L)}$ = $\frac{0.01}{50/1000}$

  = 0.212 mole/L

  =0.212 mole/dm^3

  = 0.212 M

Example (2): A bottle of dilute hydrochloric acid is 2M. What mass of hydrogen chloride, HCl would be dissolved in 100 Cm^3 of this acid ?. (M.wt HCl = (36.5)

- Solution:

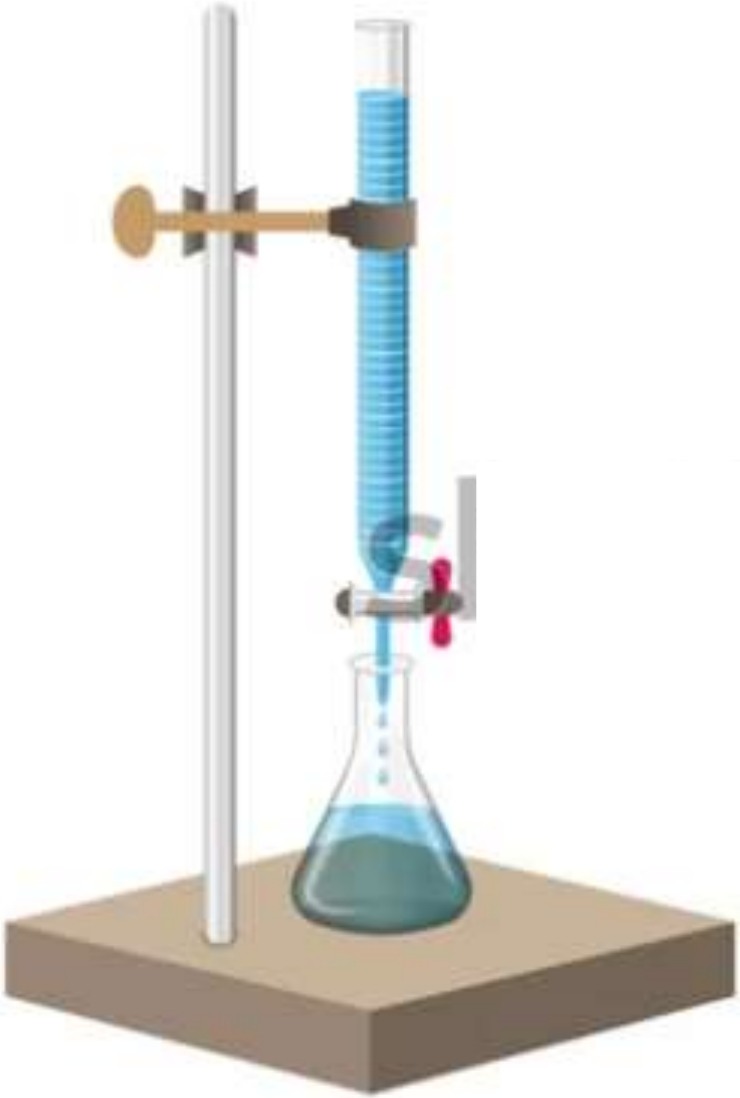
$$wt = \frac{M \times V \times M.wt}{1000}$$

$$= \frac{2 \times 100 \times 36.5}{1000} = 7.3\ g$$

Volumetric Analysis

This is where we use a known solution and react it with a solution of the chemical being tested (unknown). The typical method of doing this is in a titration.

A titration is a technique where a solution of known concentration is used to determine the concentration of an unknown solution. Typically, the **titrant** (the know solution) is added from a buret to a known quantity of the **analyte** (the unknown solution) until the reaction is complete



to get the titration law Consider the general reaction between A & B:

$$a\,A + b\,B \longrightarrow c\,C + d\,D$$

a : b

$$\frac{M_{(A)} \times V_{(A)}}{a} = \frac{M_{(B)} \times V_{(B)}}{b}$$

M_A = concentration of A

V_A = volume of A

M_B = concentration of B

V_B = volume of B

a = number of mole of A , b = number of mole of B

Neutralization (Acid-base) titrations:

The reaction between acid & base is called **neutralization reaction.**

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

acid **base**

1 : 1

$$\frac{M_{acid}V_{acid}}{a} = \frac{M_{base}V_{base}}{b}$$

where

M_{acid} = **concentration of the acid**

V_{acid} = **volume of the acid**

M_{base} = **concentration of the base**

V_{base} = **volume of the base**

mole ratio between acid and base is 1:1.

Indicators are used in titration solutions to signal the completion of the acid – base reaction

Acid-base indicators are substances which change colour with PH. They are usually weak acids or bases.Examlpes bromo phenol blue , mehyl orange, mehyl red …….

- no of moles = $\frac{M\ V}{1000}$
- (A moles) a = $\frac{Ma\ Va}{1000}$, (B moles) b= $\frac{MbVb}{1000}$
- a/b = $\frac{MaVa}{1000} / \frac{MbVb}{1000} = \frac{Va\ Ma}{Vb\ Mb}$
- $\frac{Ma\ Va}{a} = \frac{MbVb}{b}$

Example:

What volume of 0.2M NaOH solution is needed to neutralize 25ml of 0.25MHCl?

(concentration in g/l) :

It is amount of solute in grams present in one liter of solution .

$$C\,(g/L) = \frac{Wt\,(g)}{V\,(L)} \qquad \text{since } Wt = M \times M.wt \times V\,(L)$$

$$C\,(g/L) = \frac{M \times M.wt \times V\,(L)}{V\,(L)}$$

$$C\,(g/L) = M \times M.wt$$

concentration in g/l = molarity X molecular weight

Example:

What is the concentration in g/L & mole/L of 25 ml H_2SO_4 which need to neutralize 25ml of 0.1M NaOH ?

Dilution:

To save time & space in the laboratory, solutions that are routinely used are often prepared in concentrated form called stock solutions, water or any solvent is then added.

The number of moles stays the same but more water is added , so the volume increases & the molarity decreases.

M = Moles of solute/ volume

↓ ↓ ↓

Decrease constant increase

Number of moles before dilution =

Number of moles after dilution

$M_1V_1 = M_2V_2$

The process of adding more solvent to a solution is called dilution.

The molarity of stock solutions of the common concentrated acids are:

Sulphuric acid H_2SO_4 = 18M

Nitric acid HNO_3 = 16M

Hydrochloric acid HCl = 12M

Example :

A stock solution of concentrated hydrochloric acid is 12 M. To what volume must 0.5 L of this solution be diluted to prepare 2 M solution? How much water must be added to accomplish this?

Home work(4)

- 1-How much 0.600 M oxalic acid is required to neutralize 100. mL of a 0.325 M solution of aluminum hydroxide?
- $3\ H_2C_2O_4 + 2\ Al(OH)_3 \rightarrow 6\ H_2O + Al_2(C_2O_4)3$
- 2- A stock solution of concentrated hydrochloric acid is 12 M. To what volume must 0.20 L of this solution be diluted to prepare 0.50 M solution? How much water must be added to

- 3-What is the molarity of a solution that contains 390.0 g of acetic acid, CH3COOH, dissolved in enoug acetone to make 1000.0 mL of solution?

- 4- How many moles of ammonia are in 0.250 L of a 5.00 M aqueous ammonia solution?

Equivalent Weight:

is the amount of substance that are equivalent to each other in chemical reactions.

*There are two types of reactions include the definition of equivalents,

-**neutralization** reactions &

-**oxidation-reduction** reactions.

Equivalent Weight $= \frac{\text{Molecular weight}}{a} = \frac{\text{M.wt}}{a}$

a depends upon the type of reaction considered

For neutralization reactions:

The equivalent weight of an acid
is the amount of the acid that supplies one mole of hydrogen ions (H^+)

The equivalent weight of a base
is the amount of the base that supplies one mole of hydroxide ions (OH^-).

Equivalent Weight of an acid

$$= \frac{\text{M.wt}}{a}$$

a is the number **of (H^+)**

Equivalent Weight of a base

$$= \frac{\text{M.wt}}{a}$$

a is the number **of (OH^-).**

Example :

$$HCl \rightarrow H^+ + Cl^-$$

Equivalent weight of HCl equivalent. $= \frac{36.5}{1} = 36.5$ g

$$H_2SO_4 \rightarrow 2\,H^+ + SO_4^{2-}$$

Equivalent weight of H_2SO_4 equivalent. $= \frac{98}{2} = 49$ g

$$H_3PO_4 \rightarrow 3\,H^+ + PO_4^{3-}$$

Equivalent weight of H_3PO_4 equivalent. $= \frac{98}{3} = 32.67$ g

$$\mathbf{NaOH \rightarrow Na^{+} + OH^{-}}$$

Equivalent weight of NaOH $= \frac{40}{1} = 40$ g equivalent.

$$\mathbf{Ca(OH)_2 \rightarrow Ca^{2+} + 2\ OH^{-}}$$

Equivalent weight of $Ca(OH)_2 = \frac{74}{2} = 37$ g equivalent.

For oxidation-reduction reactions:

-The equivalent weight of **an oxidizing agent** is the amount of the oxidizing agent that **gains** one mole of electrons (e^-)

-The equivalent weight of a **reducing agent** is the amount of the **reducing agent** that **loses** one mole of electrons (e^-)

For redox (reduction-oxidation) **reactions**

a is the total change in oxidation number (either up or down).

Equivalent Weight of an oxidizing agent

$$= \frac{M.wt}{a}$$

a is the decrease in the oxidation number

Equivalent Weight of a reducing agent

$$= \frac{M.wt}{a}$$

a is the increase in the oxidation number

Example:

reduction of potassium permanganate in acidic medium:

$$MnO_4^- + 5e + 8H^+ \rightarrow Mn^{2+} + 4H_2O$$

(The oxidation number of Mn in MnO_4^- = +7)

Equivalent weight = $\frac{158}{5}$ = 31.6 g equivalent.

Normality (N):

- The normality (N) of a solutions is the number of equivalent weight of solute per liter of solution.

- This means 1 normal solution contains 1 equivalent of solute per liter of solution.

- Normality (N):

Normality of a solution is the number of equivalent weights of solute dissolved in one litter of soln.

$$N = \frac{\text{number of equivalent weights}}{\text{volume of solution in L}}$$

$$N = \frac{Wt \times 1000}{\text{equivalent weight} \times V\ (mL)}$$

Example:

Calculate the normality of a solution prepared by dissolving 0.15g of $Mg(OH)_2$ in 200 mL of solution.

(Atomic weights: Mg = 24.3, O = 16, H = 1)

solution

$$N = \frac{Wt \times 1000}{\text{equivalent weight} \times V\ (mL)}$$

$$\text{equivalent weight} = \frac{M.wt}{a} = \frac{58.3}{2} = 29.15\ g.eq$$

$$N = \frac{0.15 \times 1000}{29.15 \times 200} = 0.0257\ N$$

**The normality of a solution & its molarity are related by:

$$N = \boldsymbol{a}\ M$$

Example :

Calculate the normality of each the flowing :

1 *M* H_2SO_4 solution, &

0.1 *M* HCl

Solution:

$N = \boldsymbol{a}\ M$

N of H_2SO_4 = 2 × 1 = 2 N for H_2SO_4

N of HCl = 1 × 0.1 = 0.1 N for HCl.

Example(1):

Calculate the equivalent weight of phosphoric acid.(H_3PO_4).

Example(2):

A solution of H_2SO_4 acid contains 49 grams of the acid per 500 ml solution . Calculate the normality of this solution.

Molality (m): The number of moles of solute dissolved per kilo gram (kg) of solvent.

Example:

What is the molarity of a solution containing 0.46 mole of solute in 2.0 kg water?

Given:

Moles of Solute = 0.46 mole

Kilogram of Solvent = 2.0 kg

$$\text{Molality (M)} = \frac{\text{Moles of Solute}}{\text{Kilograms of Solvent}} = \frac{0.46 \text{ mole}}{2.0 \text{ kg}}$$

= 0.23 m

Example

- Calculate the Molality of the solution made by dissolving 1.0 g of phenol in 50 g of water ?.
-
- By definition (m) = no. of moles of solute

$$\frac{\text{wt (kg) of solvent}}{}$$

$$= \frac{1.0 \times 1000}{94 \times 50} = 0.2127 \text{ m}$$

Oxidation reduction reactions (redox)

- $2Mg_{(s)} + O_{2\,(g)} \longrightarrow 2Mg^{+2}O^{-2}{}_{(s)}$
- Oxidation is:

  i- the gain of oxygen.

  ii- the loss of hydrogen.

  iii- the loss of electrons.

- Reduction is:

  i/ the loss of oxygen.

  ii/ the gain of hydrogen.

  iii/ the gain of electrons.

A chemical that is used to oxidize another is an oxidizing agent.

A chemical that is used to reduce another is a reducing agent.

Oxidizing agent:

Oxidizing agent is the substance that gains electrons.
e.g. acidified potassium permanganate $KMnO_4$/ H^+

$$MnO_4^- + 5e + 8H^+ \longrightarrow Mn^{2+} + 4H_2O \quad \text{(reduction)}$$

(Violet color) (colorless)

Reducing agent:

Reducing agent is the substance that loses electrons
e.g. oxalate ion, $C_2O_4^{2-}$:

$$C_2O_4^{2-} - 2e \longrightarrow 2CO_2 \quad \text{(oxidation)}$$

Rules of assigning the oxidation number

- Rule1. The oxidation number **of an atom is zero** in a neutral substance that contains atoms of only one element. Thus, the atoms **in O_2, O_3, P_4, S_8,** and aluminum metal all have an oxidation number **of 0.**

Rule2. The oxidation number **of simple ions is equal to the charge on the ion**. The oxidation number of sodium in the **Na^+** ion is **+1**, and the oxidation number of chlorine in the **Cl^- ion is -1**

- Rule3. The oxidation number **of hydrogen is +1** when it is combined with **a *nonmetal*** as in CH_4, NH_3, H_2O, and HCl

Rule4. The oxidation number **of hydrogen is -1** when it is combined with **a *metal*** as in. LiH, NaH, CaH_2, and $LiAlH_4$.

- Rule5. Oxygen usually has an oxidation number **of -2.** Exceptions include molecules and polyatomic ions that contain O-O bonds, such as O_2, O_3, H_2O_2, and the O_2^{2-} ion

Rule6. The sum of the oxidation numbers in a neutral compound is zero.

H_2O: 2(+1) + (-2) =

- Rule7. The sum of the oxidation numbers in a polyatomic ion is equal to the charge on the ion. The oxidation number of the sulfur atom in the SO_4^{2-} ion must **be +6**, because the sum of the oxidation numbers of the atoms in this ion must **equal -2.**
- **SO_4^{2-}: (+6) + 4(-2) = -2**

Some element give more than one type of ion. Usually these are the transition elements like ion, copper and manganese. The most common one is the hydrogen ion, H^+ which is found in acidic solutions, the hydride ion H^-, is found in solid hydrides like NaH and KH, made when hydrogen reacts with powerful reducing metals.

Like wise oxygen always has a charge of -2 when it is in an ionic compound. The exception is oxygen in peroxides such as barium peroxide, BaO_2 , the O_2^{-2} ion is present. The average charge on an oxygen atom in this ion is -1.

Oxidation number of elements in covalent compound.

- What is the oxidation number of sulphur in sulphur dioxide SO_2?
- Sulphur dioxide is electrically neutral (no over all charge).

$(S) + 2(O) = 0$

$(S) + 2(-2) = 0$

$(S) = +4$

Oxidation numbers of elements in ionic groups

- Ex: what is the oxidation number of phosphorous in PO_4^{-3} ions?
- Let us assume that oxygen has its normal oxidation number.

$(P) + 4(O) = -3$

$(P) + 4(-2) = -3$

$(P) - 8 = -3$

$(P) = +5$

Assign the oxidation numbers of the undelined atoms in the following compounds.

(a) <u>Al</u> in Al_2O_3 :- 2Al + (-2*3) = 0 ,, 2Al – 6 = 0 ,, 2Al = +6
Al=+3

(b) <u>Xe</u> in XeF_4 :- Xe + (-1*4) = 0 ,, Xe – 4 = 0 ,, Xe = +4

(c) Cr in $K_2\underline{Cr}_2O_7$
:- 2Cr + (+1*2) + (-2*7) = 0,,
2Cr+2-14=0
2Cr=+12 ,,
Cr = +6

EXAMPLES:

What is the oxidation number of Cr in $CrCl_3$?

Solution: in $CrCl_3$: Cr + (-1* 3) = 0

Cr – 3 = 0

Cr = +3

The oxidation number of Cr in $CrCl_3$ is +3

Oxidation states

Chemicals like copper (II) sulphate, or iron (III) chloride. The number in brackets is the oxidation state of the element. Transition metals can exist in variety of oxidation state, it is the name which gives us a guide to the nature of the metal ion in the compound.

The key to deciding if a reaction is a redox reaction is to look for changes in oxidation number.

- Which of the following are redox reactions?

$$H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$$

$$NaOH_{(aq)} + H_2SO_{4(aq)} \longrightarrow NaHSO_{4(aq)} + H_2O_{(l)}$$

$$KCL_{(aq)} + AgNO_{3(aq)} \longrightarrow AgCl(s) + KNO_{3(aq)}$$

If we want to balance the redox equation we have to balance the number of gains and losses electrons example the reaction of potassium permanganate and ferrous sulphate in sulphuric acid as a catalyst??

Home work(5)

A- *what is the equivalent weight of the following :*

(i) $Mg(OH)_2$

(ii) H_3PO_4

(iii) H_2SO_3

B-

What is the oxidation number of Cr in $CrCl_3$?

Solution: in $CrCl_3$

C-Assign the oxidation numbers of the atoms in the following compounds

(a) Al in Al_2O_3

(b) Xe in XeF_4

(c) Cr in $K_2Cr_2O_7$

D-What is the oxidation number of chloride ion in HCl , Cl_2 , HClO

E-What is the oxidation number of sulfur in sodium tetrathionate $Na_2S_4O_6$.

D-What is the mass % , molality and mole fraction of a solution made by dissolving 4g of sodium hydroxide in 90 g of water